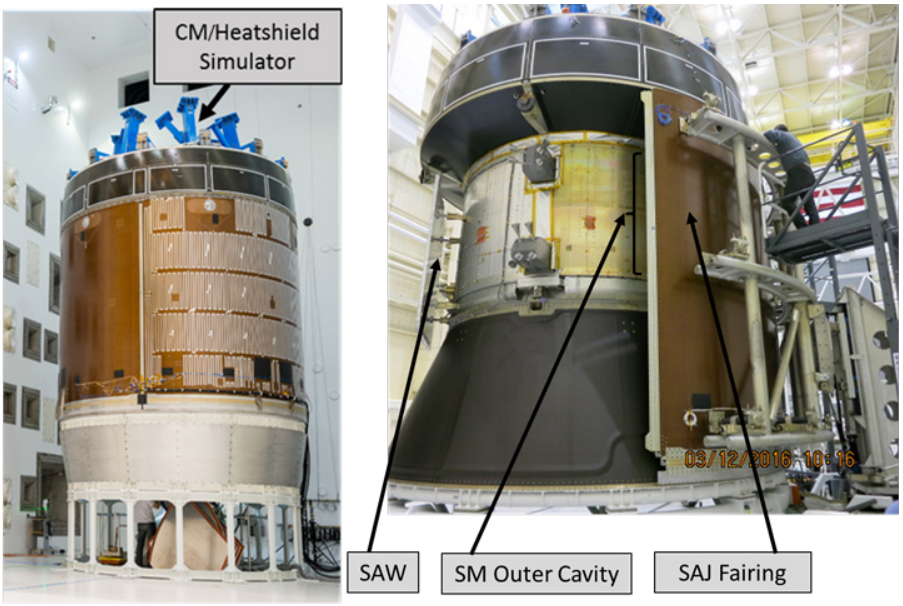
ENGINEERING CASE + WORKED EXAMPLE / ABOUT 10 MINUTES

2026-10-20 | VS&A Daily |

Same SPL, different panel loads



Orion European Service Module structural test article in its reverberant-test configuration. The solar-array wing (SAW) is acoustically sensitive. Photograph/annotations reproduced from NASA/TM-2017-219564, Fig. 2 [1]; credit: NASA and report authors. Source.

In NASA's 2016-2017 Orion test campaign, similar sound-pressure levels did not produce identical structural responses. A solar-array panel responded more strongly near 90 Hz in direct-field testing than in the reverberant test. Response limiting protected the panel [1].

This was not a perfectly controlled field-only comparison: fairings and mass simulators differed between configurations. The report explicitly explains some low-frequency differences through those changes. It is evidence that acoustic-test equivalence needs structural scrutiny, not proof that one test method is always more severe.

Today's objective is to connect microphone spectra to the force that actually excites a panel mode. We will hold every local pressure spectrum fixed and change only the relationship between locations. The resulting load can change dramatically, even though a conventional SPL overlay looks unchanged.

The key question is not only "How loud is it here?" It is also "Which parts of the structure are being pushed together, and which are being pushed against one another?"

01 / THE STRUCTURE WEIGHS THE PRESSURE FIELD

A mode does not read a microphone

Imagine a flexible panel whose two halves move in opposite directions. A pressure fluctuation that pushes both halves the same way does little work on that particular shape: one half's contribution cancels the other's. Reverse the pressure on one half and the same local pressure amplitudes now reinforce the mode. Neither picture requires a change in material, damping or natural frequency.

For a real mode shape φ_r, normalized to be dimensionless, the generalized force is the pressure weighted by normal modal displacement over the loaded area:

$$Q_r(t) = \int_A \phi_r(\mathbf{x})\, \Delta p(\mathbf{x}, t)\, \mathrm{d}A$$

Here Q_r is in newtons and Δp is the net pressure across the panel in pascals. A one-sided loading model is appropriate only when the opposite-side contribution is negligible or already included. Different mode normalization changes the numerical generalized force and modal mass together, not the physical response.

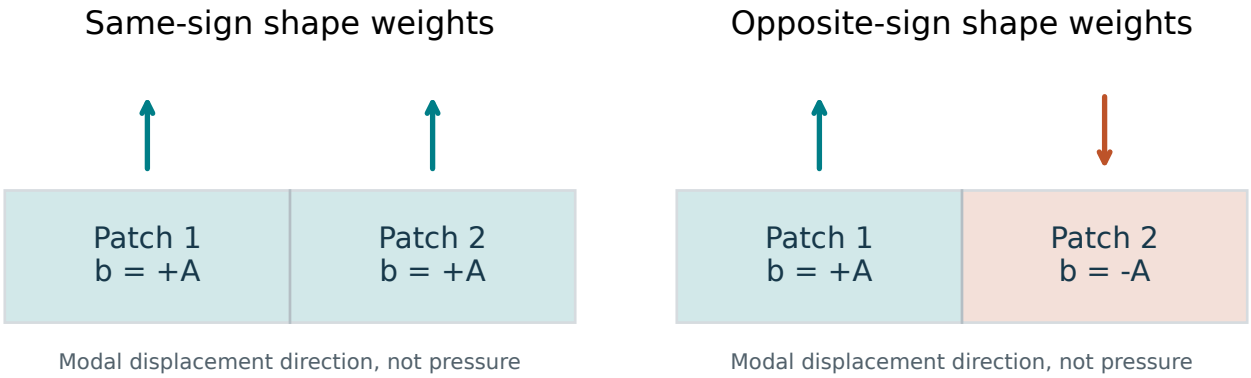


Figure 1. Original two-patch idealization. Arrows show modal motion; colors distinguish signed area weights. These are simplified shape integrals, not solved Orion modes or measured pressure maps.

Replace the integral by two uniform-pressure patches, each with effective area magnitude A. Their illustrative generalized forces are:

$$Q_+ = A(p_1 + p_2) \qquad Q_- = A(p_1 - p_2)$$

The plus and minus labels identify spatial shapes, not high and low frequencies.

This is the practical extension of source-combination physics: the signs now come from the structure. A field that cancels in one generalized coordinate can reinforce another. "Fully coherent" therefore does not identify a universally conservative loading pattern.

02 / A CONTROLLED NUMERICAL EXPERIMENT

Identical spectra, a nineteen-fold change

Assume stationary pressures with equal one-sided autospectral density $S = 400\ Pa^2/Hz$ over a 10 Hz band. Each patch has 63.25 Pa RMS in that band, or 130 dB re 20 µPa. Use $A = 0.25\ m^2$ for each signed shape weight. All values are illustrative, not Orion test data.

Define complex coherency γ using the pressure cross-spectrum S_{12}. It retains phase; magnitude-squared coherence does not:

$$\gamma = \frac{S_{12}}{\sqrt{S_{11} S_{22}}} \qquad \mathrm{MSC} = |\gamma|^2$$

$$S_{Q_\pm Q_\pm} = 2A^2 S(1 \pm \mathrm{Re}\{\gamma\})$$

Take real, frequency-independent coherency within the band. The three cases below have exactly the same individual pressure spectra. The last two columns integrate force PSD over that band.

Coherency	Force PSD (+ / -)	Force RMS (+ / -)
+0.9	95 / 5 N²/Hz	30.82 / 7.07 N
0	50 / 50 N²/Hz	22.36 / 22.36 N
-0.9	5 / 95 N²/Hz	7.07 / 30.82 N

Table 1. Original calculation. Both correlated cases have MSC = 0.81, but their cross-spectral phases differ by 180°. The pressure spectral matrices are positive definite; the example does not assert a particular loudspeaker realization.

For either fixed shape, switching between +0.9 and -0.9 changes force PSD by a factor of 19, or 12.79 dB. The corresponding force RMS ratio is √19 = 4.36. With an unchanged isolated-mode transfer function, the response PSD has the same ratio at each frequency in this band. This is not a prediction of total vehicle gRMS.

Notice the reversal between columns. Choosing the in-phase field as “worst case” protects the same-sign shape but misses the larger load on the opposite-sign shape. An incoherent model lands between these cases; it is neither a universal upper nor lower bound.

03 / FROM A TWO-PATCH SKETCH TO AN FEM

Preserve the off-diagonal terms

For an actual model, integrate the signed normal mode shape over each pressure patch. Collect those weights into a matrix B, with one row per retained mode and one column per patch. The force cross-spectral matrix follows directly:

$$\mathbf{S}_{QQ} = \mathbf{B}\,\mathbf{S}_{pp}\,\mathbf{B}^{T}$$

This expression assumes real structural modes. The pressure matrix contains local PSDs on its diagonal and complex cross-spectra elsewhere. Setting those off-diagonal terms to zero assumes independent patches; copying one random pressure signal to every patch assumes perfect in-phase coherence. Neither shortcut is neutral.

Smith and colleagues' 2012 panel/equipment study used spatially correlated pressure patches and compared predicted responses with reverberant-test measurements [2]. In an FE implementation, apply consistent distributed pressure loads or equivalent nodal forces; do not treat one pressure value as a force without its area mapping. Refine the loading discretization until the response of interest converges.

For an ideal homogeneous, isotropic three-dimensional diffuse field, the normalized point-pressure cross-spectrum takes the familiar form [2,3]:

$$\gamma(d,f) = \frac{\sin(kd)}{kd} \qquad k = \frac{2\pi f}{c}$$

Distance d is in metres, frequency f in hertz and sound speed c in m/s; the limit at zero separation is one. This is a field model, not a declaration that any reverberant chamber or speaker ring is diffuse. Boundaries, cavities, scattering and directional content can change the surface-pressure cross-spectrum substantially.

Carry the modal-force cross terms into response. With modal dynamic flexibility H, and superscript H denoting conjugate transpose:

$$\mathbf{S}_{qq} = \mathbf{H}\,\mathbf{S}_{QQ}\,\mathbf{H}^{H}$$

Reconstruct acceleration, strain or interface loads with consistent units and normalization. Nearby modes can have correlated forces; summing their response powers independently can lose that coupling.

Check structural and pressure-field discretization separately. A finer structural mesh cannot repair missing pressure variation. Adding independent patches without preserving their physical correlation changes the loading model rather than merely refining it.

04 / WHAT TO ASK BEFORE ACCEPTING THE TEST

Uniform is not the same as equivalent

Separate three questions in the review: are the local levels correct, is their spatial relationship credible, and are the critical structural responses adequately exercised? A set of microphones can have nearly equal PSDs while their phases and coherencies produce very different modal forces. Spatial uniformity and spatial correlation are distinct checks.

Plan simultaneous, phase-consistent microphone measurements over important loading regions, including both sides of a panel or adjacent cavities when differential pressure matters. Preserve complex cross-spectra and microphone coordinates. A coherence-magnitude plot alone cannot distinguish the two 0.81 cases in the worked example. Use enough averaging to assess stability, and investigate instrumentation noise before interpreting low coherence as genuine independence.

Do not restrict verification to the microphones used by the controller. Johnson and colleagues' 2018 DFAT investigation showed, in simulation, that a control-point target could be met while other field locations differed. Their analysis also emphasized source-to-microphone transfer-matrix conditioning and source configuration [3]. The engineering implication is to check monitor locations and structural response, not infer success everywhere from success at the feedback channels.

Microphone placement deserves the same skepticism as accelerometer placement. NASA-HDBK-7010 advises avoiding symmetry relative to the speaker array that could repeatedly sample an interference pattern, and recommends additional monitor microphones [4]. That is acoustic-field sampling guidance, not a universal ban on equally spaced modal-test accelerometers. A deliberate spatial mapping array can still be regular; its spacing and extent must resolve the field feature being investigated.

Use the pre-test FEM to identify pressure-sensitive modes and strain-critical regions, then compare plausible field models at fixed local spectra. If results are highly sensitive, that is a reason to improve field characterization. It is not a reason to select whichever correlation assumption produces the desired margin. Where response limiting is used, review which other components or modes may lose excitation.

The handbook was approved February 1, 2016 and revalidated January 9, 2026. It supplies DFAT guidance and does not supersede other agency requirements; contractual and program adoption determine applicability [4]. This lesson's numbers are teaching choices, not test tolerances or qualification criteria.

The decisive deliverable is a traceable chain from pressure field to structural response. Matching SPL is an important link in that chain, but it is not the entire chain.

05 / MAKE THE LOAD PATTERN VISIBLE

Three takeaways

1. A mode integrates signed pressure work. Local SPL cannot tell you whether different surface regions reinforce or cancel its generalized force.

2. Phase survives in cross-spectra, not MSC alone. In the example, the same pressure PSDs and the same MSC support a nineteen-fold change in force PSD.

3. Verify the field and the structure. Use a consistent FEM pressure mapping, independent monitor locations and critical responses; do not equate control-point compliance with field equivalence.

A short reflection

A pressure model applies one common random time history to every panel patch. Is that conservative for all modes?

Answer: No. It can strongly drive a same-sign shape while cancelling a sign-changing shape. Preserve a physically credible spatial cross-spectral matrix and test the sensitivity of the actual response quantities. There is no single correlation choice that maximizes every modal load simultaneously.

Sources and figure notes

[1] M. E. McNelis et al., Examination of the Structural Response of the Orion European Service Module to Reverberant and Direct Field Acoustic Testing, NASA/TM-2017-219564, September 2017. See pp. 3-7; opening photograph is Fig. 2.

[2] A. Smith, B. LaVerde, C. Fulcher and R. Hunt, Validation of Vehicle Panel/Equipment Response from Diffuse Acoustic Field Excitation Using Spatially Correlated Transfer Function Approach, April 23, 2012. See pp. 9-12 for pressure matrices and response formulation.

[3] A. Johnson et al., Investigation of Multi-Input, Multi-Output (MIMO) Random Control Applied to Direct Field Acoustic Testing, NESC Loads & Dynamics TDT, March 12, 2018. Slides 4, 11-14, 25-28.

[4] NASA-HDBK-7010, Direct Field Acoustic Testing (DFAT), approved February 1, 2016; revalidated January 9, 2026. Sections 1, 2.4 and 6.3.1. Guidance scope and acoustic microphone-placement recommendations.

Figure 1 and Table 1 are original illustrative calculations, not measured data or an Orion reconstruction. This run could not verify or select GPT-6 Astra.